

360° RAYS - Radial Analysis of High Dimensional Data

Bipasha Garg¹[0009-0006-8213-7300] and Kamalakara
Karlalalem¹[0000-0003-2528-7979]

Data Science and Analytics Center,
International Institute of Information Technology, Hyderabad, India
bipasha.garg@research.iiit.ac.in
kamal@iiit.ac.in

Abstract. Visualising high-dimensional real data with all dimensions while providing some semantics is a challenging problem in data analysis.

In this paper, we present the visualisation of high-dimensional real data using our implementation tool, *360° RAYS*. It is a visualisation built to plot high-dimensional data without the loss of information, unlike dimensionality reduction techniques. RAYS helps plot all the features and points in the dataset and aims to provide information such as the **geometrical positioning of points or classes in the orthants of space**, also aiding in understanding the **closeness of multiple points**, **clustering of points**, and **ordering of subspaces**, etc. This visualisation is a novel technique built after exploring major viable options to plot higher-dimensional data, which included parallel coordinates, circle segments, among others. The key idea of our approach is a **hierarchical concentric ring structure**, where each successive ring incrementally adds a dimension to the existing subspace, ordered by the dimension's variance. Data points are mapped to ring sectors based on a **bitwise encoding of their coordinate values** (positive/non-negative or negative).

The advantage of our approach is that it presents the complete data with all the dimensions and provides a **clear visual grouping of points by subspace concentration**. The visualisation also shows classes within the data. Hence, it enables the complete representation of the data in a flattened 2D radial layout.

Keywords: High Dimensional Data · Data Visualization · Radial Geometry · Subspace Analysis · Variance Ordering · Bitwise Encoding

1 Introduction

In this research paper, we study various types of data visualisation, such as **parallel coordinates** [3] and **circle segments** [1], that are used to visualise multi-dimensional data, and explain why our visualisation is different and useful in terms of visualising high-dimensional data.

The core idea of **RAYS** is to **visualise high-dimensional data while providing meaningful inferences** to the user and studying this usage of orthants (quadrants) in space. The visualisation aims to provide a **clutter-free view** of the $\mathbb{R}^d$ data, where d is the dimension of the dataset, along with a meaningful distribution of the data points and the classes (if any).

The goal of the visualisation is to help the user understand the **relative positioning** of the data points and the **usage of different orthants** in the space. This novel technique aims to provide users, typically data scientists, with an overview of the spread and geometric positions of the data points in the dataset. It makes use of the data in its raw form, i.e., without any dimensionality reduction techniques applied, providing a clearer view of the relationship between different points and orthants without the loss of any information.

We will first study how this visualisation is constructed and its structure, and how this structure serves the purposes mentioned above. While discussing the construction, we will also highlight the relevance of the visualisation in the present time.

2 RAYS

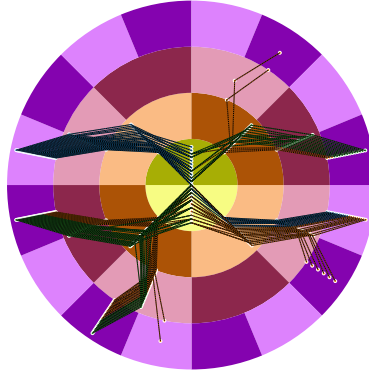


Fig. 1: Iris Dataset - RAYS Visualisation

RAYS is used to visualise high-dimensional data without the loss of information, essentially without incorporating any dimensionality reduction techniques. This novel visualisation makes use of subspaces and quadrants in which the point can lie, which helps the users study the patterns of the arrangement of points and potentially identify similarities or dissimilarities between the points.

The visualisation includes various different components which, when put together, build the complete visualisation. The visualisation aims to represent a

high-dimensional real dataset as a collection of concentric rings divided into sectors. This structure is fundamentally hierarchical and radial, which is crucial for managing the complexity and preventing the visualisation from becoming congested and cluttered. The data points are distributed across the rings and sectors, and their relationships are shown using edges.

2.1 Idea

This section introduces the core concepts and terminologies essential for understanding the construction and interpretation of the RAYS visualisation.

1. Dataset Formalisation: We introduce formal definitions of the data structure being analysed:

1. We consider an $\mathbb{R}^d$ real dataset, where $\mathbb{R}$ represents the set of all real numbers and d denotes the **dimensionality of the feature space**.
2. Let the dataset comprise n distinct data points, denoted as $x_1, x_2, x_3, \dots, x_n$, where each point x_i belongs to the $\mathbb{R}^d$ space.
3. Each data point x_i is represented by a vector of d coordinates: $(x_{i,1}, x_{i,2}, \dots, x_{i,d})$ where $x_{i,j} \in [-1, 1]$ for all $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, d\}$.
4. Let $a_1, a_2, \dots, a_d$ be the dimensions or attributes in the dataset.

2. Dimension Ordering by Variance: Without loss of generality, we **order the dimensions** $(a_1, a_2, \dots, a_d)$ based on the **increasing magnitude of their variance**.

Let $\sigma_{a_1}, \sigma_{a_2}, \dots, \sigma_{a_d}$ be the variance of the d dimensions, and

$$\sigma_{a_1} \leq \sigma_{a_2} \leq \dots \leq \sigma_{a_d}$$

This ensures that the dimensions contributing the **least** spread (variance) to the data are placed in the innermost rings, providing a stable, low-variance base for subsequent dimensions. Conversely, the dimensions contributing the **most** spread (variance) to the data are strategically placed towards the outer regions. This placement is important because, as observed in the radial geometry, the **area available for partitioning exponentially increases** with the radius. Placing high-variance dimensions in the outer rings provides a larger angular and radial area to contain the greater spread of points, thereby achieving a clearer distribution and minimizing visual clutter and **overlap** during the iterative sector subdivision process. This reduction in clutter with the help of variance can also be learnt from [2], where the author uses different techniques like pattern based clustering to reduce clutter and maximise the pattern identification in parallel coordinates.

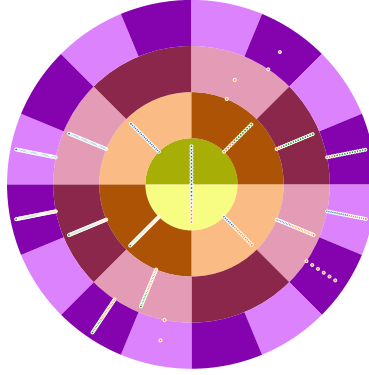


Fig. 2: Iris Dataset points plotted in hierarchical rings and sectors.

3. Hierarchical Concentric Ring Structure The RAYS visualisation is built based upon a hierarchical concentric ring structure. Each successive ring, moving outward from the centre, corresponds to the incremental addition of one dimension, following the variance-based order obtained in the previous step.

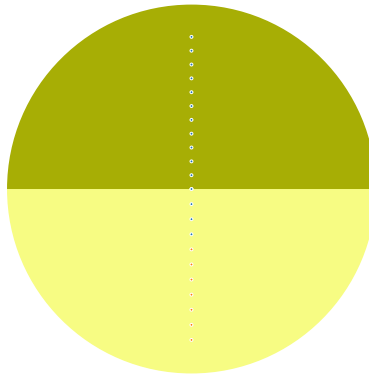


Fig. 3: 1st concentric circle of the Iris Dataset which represents SepalWidthcm attribute.

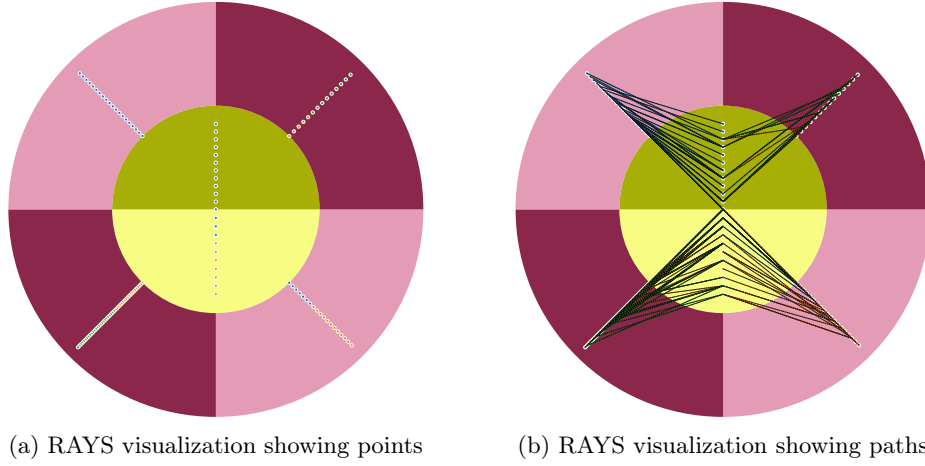


Fig. 4: Two-Ring RAYS Visualization for Iris Dataset: (a) depicts the distribution of individual data points within the concentric ring structure, while (b) illustrates the continuous paths (edges) formed by these points across the rings, representing their full coordinate sequences up to the second dimension.

- **Innermost Circle (Dimension a_1):** The innermost circle plots all points' first coordinate, $x_{i,1}$ (where $1 \leq i \leq n$). This circle is fundamentally divided into two major sectors (semi-circles): one for the positive/non-negative values ($x_{i,1} \geq 0$) and one for the negative values ($x_{i,1} < 0$), as shown in Figure 3. The darker region represents the positive/non-negative sector and the lighter region represents the negative sector.
- **Successive Rings (Dimension a_j):** The next concentric ring, representing the second dimension a_2 , plots the coordinate $x_{i,2}$. The location of $x_{i,2}$ is **relative** to the sector established by $x_{i,1}$, and is further subdivided into two new sub-sectors based on the sign (positive or negative) of $x_{i,2}$ as shown in Figure 4a.
- **Recursive Partitioning:** This process is repeated for every dimension $j = 1$ through d . Each ring j adds the dimension a_j , recursively partitioning the sectors established by the previous $j - 1$ dimensions based on the sign of the j^{th} coordinate ($x_{i,j}$). After d dimensions, the total number of distinct angular sectors created is 2^d , corresponding precisely to the 2^d orthants in the $\mathbb{R}^d$ space. The final visualization obtained after plotting the points in d rings will be like for example: Figure 2. The color of the points represent different labels.

4. Point Representation A point's location, up to the $j - 1$ dimension, ($x_{i,1}, \dots, x_{i,j-1}$), is represented by a path that traverses from the sector corresponding to $x_{i,1}$, to $x_{i,2}$, and so on, up to $x_{i,j-1}$ across the $j - 1$ dimensions (rings) as shown in Figure 4b.

Once the j^{th} dimension is added, the path from $x_{i,j-1}$ to $x_{i,j}$ proceeds based on the value (positive or negative) of the j^{th} coordinate.

The final representation of the complete d -dimensional point $x_i = (x_{i,1}, \dots, x_{i,d})$ is the **entire continuous path** extending from the innermost concentric circle (Dimension a_1) to the outermost concentric ring (Dimension a_d), as illustrated in Figure 1. This path's angular extent uniquely identifies **orthant** to which the point x_i belongs in the $\mathbb{R}^d$ space.

2.2 Algorithm

Using the terminology defined previously, we apply the following algorithm to set up the **RAYs** visualisation structure:

1. **Input:** $\mathbf{X}$, a d -dimensional dataset with n points, $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$.
2. **Point Format:** Each point x_i is represented as a vector: $(x_{i,1}, x_{i,2}, \dots, x_{i,d})$ where, $x_{i,j} \in [-1, 1]$ for $1 \leq i \leq n$ and $1 \leq j \leq d$ (after normalisation).
3. **Dimension Ordering:** The indices $j = 1, \dots, d$ correspond to the dimensions $a_1, a_2, \dots, a_d$, which are ordered by non-decreasing variance:

$$\sigma_{a_1} \leq \sigma_{a_2} \leq \dots \leq \sigma_{a_d}$$

Algorithm 1 RAYs Visualisation Construction

- 1: **procedure** CONSTRUCTRAYs($\mathbf{X}$)
 - 2: **Initialisation (Dimension a_1):**
 - 3: Create the innermost concentric circle with radius r_1 .
 - 4: Partition the circle into two sectors based on the sign of $x_{i,1}$:
 - 5: • One sector for $x_{i,1} \geq 0$ (e.g., the positive half).
 - 6: • One sector for $x_{i,1} < 0$ (e.g., negative half).
 - 7: Place all points' first coordinates $x_{i,1}$ into the corresponding sectors of the innermost circle.
 - 8: **for** $j = 2$ **to** d **do**
 - 9: **Iterative Partitioning (Dimension a_j):**
 - 10: Create the j^{th} concentric ring (radius r_j).
 - 11: **for** each existing path segment ending at dimension a_{j-1} in ring $j - 1$ **do**
 - 12: Subdivide the corresponding sector in ring j into two new sub-sectors.
 - 13: The subdivision is based on the sign of the current coordinate $x_{i,j}$:
 - 14: • One for $x_{i,j} \geq 0$.
 - 15: • One for $x_{i,j} < 0$.
 - 16: Place the point and connect the point's path in continuation into the correct sub-sector based on $x_{i,j}$.
 - 17: **Output:** The points are located in its orthant/sector relative to its previous dimension's point position. A radial layout where each point x_i is represented by a unique continuous path from r_1 to r_d , identifying its orthant/sector based on the bitwise sign encoding of its coordinates $(x_{i,1}, \dots, x_{i,d})$.
-

2.3 Properties

The **RAYS** visualisation exhibits the following key properties, which significantly aid the user in identifying underlying patterns and structural characteristics within the dataset:

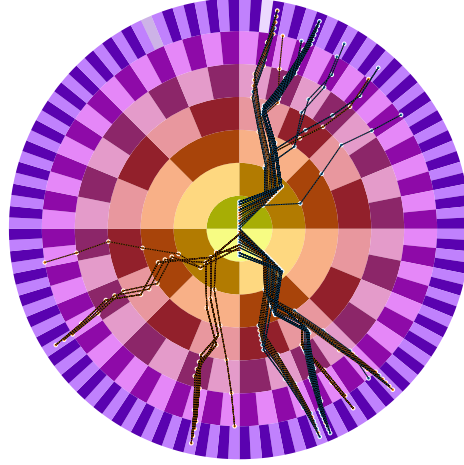


Fig. 5: Sleep-lifestyle (8-dimensional) Dataset with 2 labels - RAYS Visualisation

- **Spread and Sparsity (Orthant Usage):** As the number of dimensions (d) increases, the $\mathbb{R}^d$ space becomes increasingly sparse. The number of orthants in $\mathbb{R}^d$ space is 2^d , growing exponentially with d . Since the number of data points (n) typically does not grow at the same exponential rate, many orthants are left entirely unused as we can observe in Figure 5. The RAYS visualisation depicts this **usability of the sectors/orthants**, allowing analysts to understand the **spread** and geometric concentration of the data across the high-dimensional space.
- **Continuity and Clutter Reduction:** The visualisation leverages the **hierarchical nature of the sectors**, where the angular region defined by a parent sector fully includes the combined region of its child sectors. This hierarchical structure ensures that the paths representing individual points possess **continuity**. Furthermore, the edges connecting successive coordinates of a single point are assured **never to criss-cross** the edges of any other point within the various sector linkages. This feature minimises visual clutter, ensuring that each point x_i is clearly represented by a **unique continuous path** from the innermost circle (r_1) to the outermost ring (r_d), where $1 \leq i \leq n$.

- **Commonality Grouping:** If the input dataset $\mathbf{X}$ is a classified dataset (i.e., has class labels) for example in Figure 5, RAYS offers a method to analyse the **commonality and purity of sectors/orthants**. For example: if a sector S_j within a specific ring R_i contains point coordinates belonging exclusively to one class, we define that sector as **pure** as in Figure 5, sectors in the outermost ring containing only orange or only blue points. Conversely, if the sector contains a mixture of various classes, we define it as **mixed**. This analysis allows the user to quantify how many sectors are pure versus mixed, providing a data analyst with insights into the general **geometric distribution of classes** in the dataset.

2.4 Time and Space Complexity

The analysis of the **RAYS** visualisation setup focuses on the number of data points (n) and the dimensionality of the dataset (d).

Time Complexity The major operation in the RAYS setup is the processing and placement of each coordinate for every data point, as described in Algorithm 1.

- The algorithm iterates through all n points.
- For each point x_i , it iterates through all d dimensions (the loop for $j = 2$ to d and the initial step for $j = 1$).
- The core operation within the loops involves reading the coordinate value $x_{i,j}$ and mapping the point's path to the correct sector based on the bit-wise sign encoding. This mapping (sector subdivision) is performed once per coordinate.

Therefore, the total number of operations required is proportional to the total number of coordinates in the dataset, which is $n \times d$.

The time complexity for setting up the RAYS visualisation is $\mathbf{O(nd)}$.

Space Complexity The space required is primarily determined by the need to store the input data and the structure representing the radial layout.

- **Data Storage:** The input dataset $\mathbf{X}$ requires space proportional to $n \times d$ to store all coordinates.
- **Structure Storage:** The RAYS output represents each of the n points as a path that traverses d rings. Storing these n paths, each having d segments, requires space proportional to $n \times d$.

Since the space required to store the input and the space required to store the resulting paths are both proportional to the size of the input, the space complexity is also $\mathbf{O(nd)}$.

3 Visual Aspects

The **RAYS** visualisation is designed to provide specific visual cues that facilitate pattern recognition and data interpretation:

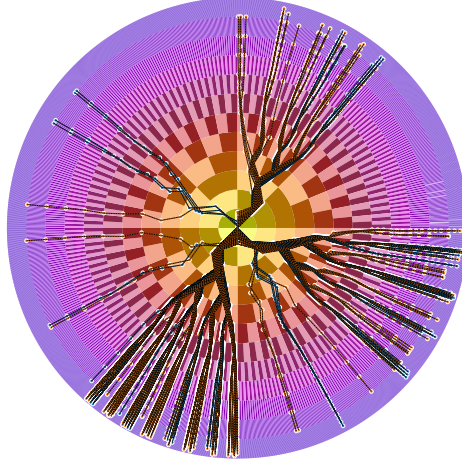


Fig. 6: Tsunami (12-dimensional) Dataset with 2 labels - RAYS Visualisation

- **Orthant Usage:** The visualisation directly facilitates the identification of highly used sectors versus those that remain empty, meaning no data point belongs to the corresponding high-dimensional orthant in the $\mathbb{R}^d$ space as in Figure 6, we observe the high number of empty orthants/sectors. Due to the sparsity of high-dimensional data, many such unused sectors can be readily identified. This capability provides advantage over techniques like **Circle Segments** [1], which primarily provide a density-based overview using pixel-based representations, often obscuring the underlying geometric structure.
- **Point Identification and Cognitive Load Reduction:** Identifying and tracing a specific data point based on its position is simplified due to the **hierarchical nature of the visualisation**. Unlike **Parallel Coordinates**, where understanding a complete data point requires visually tracing a line across d separate axes (increasing cognitive load), RAYS represents the point as a single, continuous, angular path. This hierarchical structure reduces the cognitive load by storing the information of previous coordinates within the current sector's position. Furthermore, the bitwise encoding embedded in each sector allows the user to identify the point's orthant (sign pattern) at any given ring without needing to recall prior coordinate details.

- **Integrated Class Semantics:** RAYS effectively integrates class information directly into the visual structure. By using distinct colours or visual properties for different classes, the visualisation reveals the geometric positions of these classes within the defined sectors/orthants.

4 Datasets and Analysis

Following the development of the visualisation technique, its effectiveness was tested and observed on various high-dimensional datasets.

Data Format and Preparation: The raw data was initially acquired in the CSV format. For better usability, particularly for rendering via web-based libraries like D3.js, the datasets were converted into the JSON format.

Public Repository and Results: All datasets used for testing and validation, along with the corresponding output visualisations and detailed analytical results, are made publicly available.

The resources can be accessed at the following public repository:

<https://github.com/Bipasha-Garg/PAKDD-Data-Results.git>

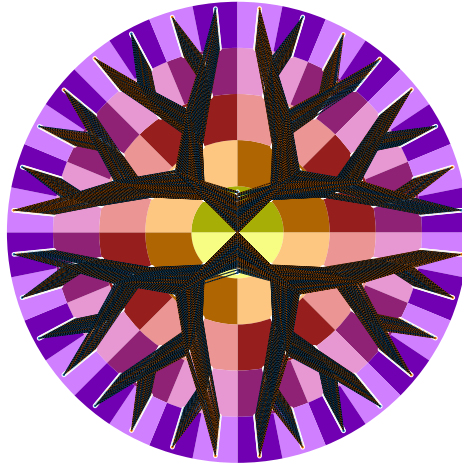


Fig. 7: Hypertension Identification (5-dimensional) Dataset with 2 labels - RAYS Visualisation

5 Limitations

The developed visualisation provides an intuitive and hierarchical method for the study and analysis of high-dimensional data, leveraging radial symmetry and

bitwise encoding to manage complexity. However, the current implementation incurs certain computational overheads and limitations.

The primary limitations of the current visualisation method stem from resource usage:

- **Dataset Requirements:** In the case of a **highly uniform dataset** combined with a **large number of dimensions** (d), all 2^d sectors of the outer rings will tend to be utilized. This high utilization dramatically increases the complexity of the visualization, causing a need for **more computational resources** to render and process the path connections. Furthermore, due to the data’s uniformity and the dense occupation of orthants, the key benefit of RAYS—identifying useful geometric patterns like empty orthants or class-pure sectors—is significantly diminished, as illustrated in Figure 7.
- **Computational Cost and Scale:** The visualisation becomes computationally demanding when dealing with extremely large datasets (high n) or very high dimensionality (high d).
- **Storage Overhead:** The current practice of storing the dataset and its associated structural metadata (especially for sectors and edges) as a JSON file can lead to significant **space complexity**. This file size directly impacts load times.
- **Rendering Performance (D3.js):** The rendering of the visualisation using the **D3.js library** can be memory-intensive. Drawing a large number of individual points and edges, especially in the outer rings with 2^d sectors, requires large memory and processing power for smooth interactive display. The complexity, $O(n \cdot d)$ for points and edges, grows linearly with both the number of points and dimensions, making rendering a bottleneck for large n and d .
- **Angular Space reduction:** As the dimension i increases, the angular width of sectors decreases exponentially ($\Delta\theta_i = 360^\circ/2^i$). While the increased radius compensates for this, the visualisation can still become overly dense and difficult to interpret visually in the outermost rings (R_d for large d).

6 Conclusion

The visualization of high-dimensional data remains a significant challenge in data analysis. While a large number of techniques exist to plot such data, the **RAYS** visualization presents a novel solution by directly mapping the **raw, high-dimensional data** into a flattened 2D radial layout, thereby completely preserving the original information without relying on dimensionality reduction.

The primary contribution of RAYS lies in its ability to visually outline meaningful structural information, such as orthant space usage and the geometric distribution of data points and classes. Its hierarchical, variance-ordered concentric ring structure effectively translates the complexity of $\mathbb{R}^d$ space into an intuitive format that helps in the identification of clusters and sparse regions.

Although RAYS has inherent limitations, particularly its reduced efficacy when dealing with extremely uniform, high-dimensional datasets, it provides

crucial information to data analysts in an easily understandable method. By providing a clear and comprehensive representation of the complete dataset in a single view, RAYS enables the naked eye to grasp the fundamental spread, sparsity, and class commonality of the dataset at a glance.

7 AI Use Statement

Generative AI tools were used to assist with grammar checking, sentence refinement, and improving clarity of technical explanations in this manuscript. All scientific content, methodology, analysis, and conclusions are the work of the author. The AI was not used to generate scientific claims or interpret results.

References

1. Ankerst, Mihael Keim, Daniel Kröger, Peer. (1996). Circle Segments : A Technique for Visually Exploring Large Multidimensional Data Sets. First publ. in: Visualization '96, Hot Topic Session, San Francisco, CA, November, 1996.
2. Hemant Makwana, Sanjay Tanwani, Suresh Jain . Axes Re-Ordering in Parallel Coordinate for Pattern Optimization. International Journal of Computer Applications. 40, 13 (February 2012), 43-48. DOI=10.5120/5044-7370
3. Johansson Westberg, Jimmy Forsell, Camilla. (2015). Evaluation of Parallel Coordinates: Overview, Categorization and Guidelines for Future Research. IEEE Transactions on Visualization and Computer Graphics. 22. 1-1. 10.1109/TVCG.2015.2466992.